

Article

Reliability-Aware Multi-Modal Sentiment Analysis Under Missing and Corrupted Modalities

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Abstract

Multi-modal sentiment analysis integrates linguistic, acoustic, and visual evidence, yet the reliability of these streams varies across samples because of missing observations, masking, measurement noise, and feature corruption. This paper presents a trainable reliability-aware evidential fusion framework that estimates not only sentiment predictions but also modality-specific evidence, predictive uncertainty, observable input quality, cross-modal disagreement, and normalized sample-dependent fusion weights. Each available modality is independently encoded and processed by an evidential classification head and a quality estimation head. Availability masks enforce exact exclusion of missing streams, while estimated quality, Dirichlet uncertainty, and Jensen–Shannon disagreement jointly regulate the contribution of each observed stream. The model is optimized end-to-end using fused classification, evidential regularization, clean–corrupted consistency, reliability-calibrated cross-modal alignment, and quality regression objectives. Experiments are conducted on both CMU-MOSI and CMU-MOSEI using their official speaker-independent splits. Binary classification follows the standard non-zero protocol, in which samples with sentiment score zero are excluded from Acc-2 and binary F1 evaluation; all labeled samples are retained for seven-class accuracy, mean absolute error, and correlation. The evaluation covers complete-input, every single- and double-modality missing pattern, graded and unseen corruption, combined missing-plus-corrupted conditions, calibration, selective prediction, statistical testing, and computational efficiency. All comparative values in the main tables are identified as local controlled adaptations under the common pipeline, while selected published reference values are reported separately to prevent provenance mixing. Across both datasets, the empirical results show that the proposed method preserves competitive complete-input performance while providing larger and more consistent gains as modality availability or integrity deteriorates.



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Keywords: multimodal sentiment analysis; uncertainty estimation; reliability-aware fusion; missing modality; corrupted modality; calibration; CMU-MOSI; CMU-MOSEI

1. Introduction

Multi-modal sentiment analysis (MSA) aims to infer affective polarity by jointly modeling linguistic content, vocal behavior, and facial activity in spoken videos. Language usually carries explicit semantic information, whereas acoustic prosody and visual

expression provide complementary cues such as emphasis, hesitation, facial action, and incongruity. Public resources such as CMU-MOSI and CMU-MOSEI have made this setting a standard benchmark for multi-modal language understanding [1,2]. Representative methods include tensor-based interaction modeling [3], low-rank fusion [4], cross-modal attention [5], modality-invariant and modality-specific learning [6], self-supervised unimodal supervision [7], and mutual information maximization [8,9].

Most of these methods optimize predictive accuracy under the assumption that all modalities are present and sufficiently clean. This assumption is fragile. Automatic transcripts can contain deletions, acoustic features can be disturbed by background noise, visual features can be partially missing because of occlusion or tracking failure, and an entire modality can be unavailable. Static fusion may continue to assign a large contribution to an unreliable modality and may produce a confident but incorrect prediction. Prior work has therefore studied modality robustness, uncertain missing modalities, dynamic fusion, and evidential multi-view learning [10–15].

A reliability-aware idea is useful only if the uncertainty and reliability values can be computed in an actual forward pass. We therefore formulate the framework as an end-to-end neural model rather than as a conceptual checklist. For every observed stream, an evidence head produces a Dirichlet distribution and an uncertainty value; a separate quality head predicts input integrity from the encoded stream. During training, the quality head receives continuous targets derived from known synthetic corruption severity. During inference, no corruption label is required: the quality score, evidence, uncertainty, disagreement, and availability mask are all computed from the current sample. Missing streams are excluded exactly through the availability mask, and the remaining weights are renormalized.

The study addresses six prespecified questions. **RQ1:** Does reliability-aware evidential fusion preserve complete-input performance on both CMU-MOSI and CMU-MOSEI? **RQ2:** Does the complete framework improve robustness across every single- and double-modality missing pattern? **RQ3:** Do the learned quality and reliability signals improve performance under graded, heterogeneous mixed, and previously unseen corruption? **RQ4:** Do evidential supervision and reliability-calibrated alignment improve calibration and selective prediction? **RQ5:** Are the robustness gains maintained across random seeds and disproportionate to the added computational cost? **RQ6:** Are baseline provenance, custom metrics, and direct quality diagnostics sufficiently explicit to support an auditable interpretation of the comparisons? These questions determine the organization of the experimental protocol and results.

The contributions are fourfold. First, we formulate a complete reliability-aware evidential fusion model in which availability, observable quality, predictive uncertainty, and cross-modal disagreement have distinct mathematical roles and are jointly optimized in a single forward path. Second, we introduce corruption-supervised quality estimation and reliability-calibrated representation alignment, allowing the model to reduce the influence of degraded streams without treating every disagreement as an error. Third, we establish a unified two-dataset protocol on CMU-MOSI and CMU-MOSEI with a consistent non-zero binary classification definition, official speaker-independent splits, fixed missingness patterns, graded corruption, unseen operators, calibration, selective prediction, and efficiency evaluation. Fourth, we provide mechanism-level and loss-level ablations, paired statistical comparisons, calibration and selective-prediction analyses, direct quality diagnostics, computational-cost controls, and an explicit published-versus-local baseline audit to connect each claim to a directly evaluated component.

The remainder of the paper is organized as follows. Section 2 reviews multi-modal sentiment fusion, uncertainty-aware fusion, and missing modality learning. Section 3

defines the problem. Section 4 presents the trainable reliability-aware method. Section 5 details the two benchmarks and unified experimental protocol. Section 6 reports the experiments. Section 7 discusses limitations, and Section 8 concludes the paper.

2. Related Work

2.1. Multi-Modal Sentiment Fusion

Early MSA systems concatenated hand-crafted or learned unimodal features. Tensor Fusion Network models unimodal and cross-modal interactions through an outer-product tensor, whereas Low-rank Multi-modal Fusion reduces its computational cost [3,4,16]. MulT uses directional cross-modal attention to model unaligned sequences [5]. MISA separates modality-invariant and modality-specific subspaces [6]; Self-MM introduces unimodal auxiliary supervision [7]; and MMIM maximizes mutual information within and across modalities [8,17]. These methods establish strong representation learning baselines, but they do not by themselves guarantee calibrated behavior when modality quality changes.

2.2. Uncertainty and Trustworthy Fusion

Uncertainty estimation can be approximated with Monte Carlo dropout, deep ensembles, or explicit evidence distributions [18–20]. Evidential deep learning maps non-negative evidence to Dirichlet parameters using $\alpha_k = e_k + 1$ [20]. Trusted Multi-View Classification integrates view-specific evidence and uncertainty [15]. Multi-modal Dynamics estimates feature- and modality-level informativeness for dynamic fusion [14]. QMF provides a theoretical and empirical connection between uncertainty and dynamic multi-modal fusion [13].

Our method uses these established principles and does not claim Dirichlet evidence or inverse-uncertainty weighting as independently new. The specific contribution is an implementation-level integration for MSA: a learned observable quality score is combined with availability, evidential uncertainty, and disagreement; the quality estimator is supervised by corruption severity; and the same capacity-matched network is evaluated under complete, missing, and corrupted inputs. This positioning distinguishes the proposed model from a tutorial or perspective paper.

2.3. Missing and Corrupted Modalities

Robustness studies have shown that MSA systems can become overly dependent on a dominant modality [10,21,22]. TATE introduces missing-type tags for uncertain missing modalities [11], while MissModal aligns complete and incomplete representations through multiple consistency constraints [12]. We use controlled missing-at-random masks and do not claim to identify missing-not-at-random mechanisms. Corruptions are applied independently to language, acoustic, and visual features, enabling a direct test of whether the fusion weights decrease when the affected modality becomes less trustworthy. Recent learning-based approaches in other multi-modal domains, including medical imaging and physiological signal analysis, have also investigated robust representation learning under heterogeneous inputs [23–25]. Although these tasks differ from sentiment analysis, their emphasis on reliable feature extraction provides complementary insights.

3. Problem Formulation

Let $\mathcal{D} \in \{\text{MOSI}, \text{MOSEI}\}$ denote one of the two multi-modal sentiment datasets and let $\mathcal{M} = \{T, A, V\}$ denote the language, acoustic, and visual modalities. For sample i , the aligned input of modality m is $\mathbf{X}_i^m \in \mathbb{R}^{L_i \times d_m}$, where L_i is the number of aligned word-level steps and d_m is the modality-specific feature dimension. The original sentiment annotation is $s_i \in [-3, 3]$.

For binary evaluation, we adopt one protocol throughout the paper:

$$y_i = \mathbb{I}[s_i > 0], \quad i \in \mathcal{I}_{\text{nz}} = \{i : s_i \neq 0\}. \quad (1)$$

Thus, zero-valued samples are excluded only from Acc-2 and binary F1. For Acc-7, mean absolute error, and Pearson correlation, all samples are retained and the original sentiment target is used. This separation prevents binary and regression metrics from being computed over incompatible sample sets. For seven-class sentiment classification, the same reliability-weighted fusion mechanism is retained, while the evidence head is extended to $K = 7$ classes. For continuous sentiment prediction, an additional regression head is applied to the fused representation and optimized using mean squared error loss. These two downstream tasks are evaluated independently rather than jointly optimized with the binary classification objective. Specifically, the reliability estimation modules and fusion strategy remain unchanged, while only the task-specific prediction head is replaced according to the target output space.

Availability is represented by $a_i^m \in \{0, 1\}$. A value of zero means that the stream is structurally unavailable rather than merely numerically close to zero. A corruption operator $\mathcal{C}_{r,c}^m$ transforms an available modality under corruption family c and normalized severity $r \in [0, 1]$. The observed stream is

$$\tilde{\mathbf{X}}_i^m = a_i^m \mathcal{C}_{r_i^m, c_i^m}^m(\mathbf{X}_i^m). \quad (2)$$

When $a_i^m = 0$, its temporal mask, evidence, quality, and unnormalized reliability are set to zero, so it cannot contribute to the fused prediction.

The learning objective is to estimate

$$F_\theta : \{(\tilde{\mathbf{X}}_i^m, a_i^m)\}_{m \in \mathcal{M}} \mapsto (\mathbf{p}_i, u_i, \mathbf{w}_i, \mathbf{q}_i), \quad (3)$$

where $\mathbf{p}_i$ is the fused class distribution, u_i is fused evidential uncertainty, $\mathbf{w}_i$ contains normalized modality weights, and $\mathbf{q}_i$ contains estimated modality qualities. At least one modality is available in every evaluated condition. If all modalities are absent, the model returns an abstention state because no evidence-supported sentiment decision can be formed.

4. Trainable Reliability-Aware Evidential Fusion

4.1. Design Principles and Information Separation

The proposed method separates four concepts that are often conflated in dynamic fusion. *Availability* is a deterministic observation-state variable and is never inferred from feature magnitude. *Observable quality* describes whether an available representation resembles a clean, information-preserving input. *Predictive uncertainty* describes the amount of class evidence supplied by a modality, which can remain high even for a visually or acoustically clean but sentiment-ambiguous sample. *Disagreement* measures inconsistency between an available modality and the current multi-modal consensus. Keeping these quantities distinct prevents a missing stream from receiving a learned weight, prevents uncertainty from being interpreted as corruption severity, and prevents minority but correct evidence from being rejected solely because it disagrees with a dominant stream.

For each modality, the network therefore contains a sequence encoder, an evidence head, and a quality head. The evidence head determines the class-specific support and evidential uncertainty. The quality head estimates input integrity from the representation itself. These outputs are combined only after availability masking. The resulting reliability weights are sample-dependent, sum to one over available modalities, and remain differentiable, enabling the classification objective to shape the complete weighting mechanism.

4.2. Availability-Aware Sequence Encoding

Each modality is first transformed by a modality-specific affine projection with layer normalization and dropout and is then encoded by a bidirectional GRU. The text encoder uses 128 hidden units per direction, whereas the acoustic and visual encoders use 64 hidden units per direction. The modality-specific encoder outputs are subsequently projected to a shared 128-dimensional latent space. The projection and recurrent parameters are not shared because linguistic, acoustic, and visual streams have different temporal statistics. After this projection, the available modality sequences are refined by the cross-modal attention blocks specified in Section 5. Let $\mathbf{Z}_i^m \in \mathbb{R}^{L \times 128}$ denote the resulting interaction-enhanced padded sequence representation, and let $\mu_i^m \in \{0, 1\}^L$ denote its valid-time-step mask. The fixed-length representation is masked mean pooling

$$\mathbf{h}_i^m = \frac{\sum_{t=1}^L \mu_{it}^m \mathbf{Z}_{it}^m}{\sum_{t=1}^L \mu_{it}^m + \epsilon}. \quad (4)$$

For an unavailable modality, $\mu_i^m = \mathbf{0}$ and $\mathbf{h}_i^m = \mathbf{0}$. This explicit mask prevents zero padding from being interpreted as valid evidence.

4.3. Unimodal Evidence and Predictive Uncertainty

Each encoder is followed by an evidence head. For $K = 2$ classes, the non-negative evidence is

$$e_{ik}^m = \text{softplus}(g_m(\mathbf{h}_i^m)_k), \quad \alpha_{ik}^m = e_{ik}^m + 1. \quad (5)$$

The Dirichlet strength, class probability, and total uncertainty are

$$S_i^m = \sum_{k=1}^K \alpha_{ik}^m, \quad p_{ik}^m = \frac{\alpha_{ik}^m}{S_i^m}, \quad u_i^m = \frac{K}{S_i^m}. \quad (6)$$

Thus $u_i^m \in (0, 1]$: low evidence produces high uncertainty, whereas concentrated evidence produces low uncertainty [20].

4.4. Observable Quality Estimation

Predictive uncertainty alone cannot distinguish every low-quality input, so each stream also has a quality head

$$q_i^m = \sigma(\mathbf{v}_m^\top \text{ReLU}(\mathbf{W}_m \mathbf{h}_i^m + \mathbf{b}_m) + c_m), \quad q_i^m \in (0, 1). \quad (7)$$

During training, an observed stream is either left clean or corrupted at severity r_i^m . Its supervision target is $t_i^m = 1 - r_i^m$, with $t_i^m = 1$ for a clean stream. The quality loss is

$$\mathcal{L}_q = \frac{1}{\sum_{i,m} a_i^m} \sum_{i,m} a_i^m (q_i^m - t_i^m)^2. \quad (8)$$

The target is used only for training. At validation and test time, q_i^m is predicted directly from the observed representation, so the method does not require knowledge of the corruption type or severity. For a missing stream, q_i^m is set to zero after the forward pass. However, since the quality targets are derived from synthetic corruption severity during training, a potential distribution gap may exist between synthetic degradations and naturally occurring real-world distortions. Developing self-supervised or naturally supervised quality estimation strategies is an interesting direction for future work.

4.5. Reliability Weighting and Disagreement

A raw reliability value combines availability, quality, and uncertainty:

$$\rho_i^m = a_i^m(q_i^m + \epsilon) \exp(-\beta u_i^m). \quad (9)$$

For two or more available modalities, let $\bar{\mathbf{p}}_i$ be the average available unimodal distribution and define

$$d_i^m = \text{JS}(\mathbf{p}_i^m \parallel \bar{\mathbf{p}}_i). \quad (10)$$

If only one modality is available, $d_i^m = 0$. The normalized weight is

$$w_i^m = \frac{\rho_i^m \exp(-\gamma d_i^m)}{\sum_{n \in \mathcal{M}} \rho_i^n \exp(-\gamma d_i^n)}, \quad \sum_m w_i^m = 1. \quad (11)$$

The disagreement term is therefore not used as a stand-alone error detector. It reduces a modality's influence only in combination with its predicted quality and uncertainty.

4.6. Normalization, Boundary Cases, and Numerical Stability

The availability factor makes the reliability of a missing modality exactly zero before normalization. In the theoretical formulation, the reliability weights are normalized only over the available modalities. In implementation, a small ϵ is added to the denominator only as a numerical safeguard against unstable division and is not treated as trainable evidence. When exactly one modality is available, its normalized weight is set to one and disagreement is set to zero. When two or three streams are available, the denominator contains only their non-zero reliability scores. The exponential uncertainty and disagreement penalties are clipped to numerically stable ranges before exponentiation. Therefore, the fusion rule does not assign contribution to structurally missing modalities and remains independent of the number of unavailable modalities.

The quality and uncertainty factors are complementary. A clean but sentiment-neutral segment may have high quality and high uncertainty; a severely corrupted segment may have low quality even when its classifier is spuriously confident. Their multiplicative combination penalizes either failure mode. The disagreement factor is applied last, so disagreement cannot override an unavailable mask and cannot independently elevate an otherwise unreliable modality.

4.7. Evidence Fusion and Prediction

The fused evidence and Dirichlet parameters are

$$\mathbf{e}_i = \sum_{m \in \mathcal{M}} w_i^m \mathbf{e}_i^m, \quad \boldsymbol{\alpha}_i = \mathbf{e}_i + \mathbf{1}, \quad p_{ik} = \frac{\alpha_{ik}}{\sum_j \alpha_{ij}}. \quad (12)$$

The class output is $\hat{y}_i = \arg \max_k p_{ik}$ and fused uncertainty is $u_i = K / \sum_k \alpha_{ik}$. Figure 1 shows the exact training and inference paths. The convex aggregation is adopted because the modality weights represent normalized reliability contributions rather than independent evidence sources. This design prevents excessive evidence accumulation caused by multiple agreeing but redundant modalities and maintains a comparable uncertainty scale across different modality availability patterns. Alternative strategies such as additive evidence accumulation may increase total evidence strength when multiple modalities agree, but may also produce overconfident predictions without explicit reliability calibration.

4.8. Decision Rule and Selective Prediction

For ordinary classification, the predicted class is the maximum-probability class. For selective prediction, confidence is defined as $1 - u_i$ and a validation-selected threshold

τ determines whether the model predicts or abstains. Coverage is the fraction of samples with $1 - u_i \geq \tau$, and selective risk is the error rate on that retained subset. The threshold is selected without access to test labels. This formulation evaluates whether evidential uncertainty ranks difficult or degraded samples usefully rather than merely improving accuracy at a single operating point.

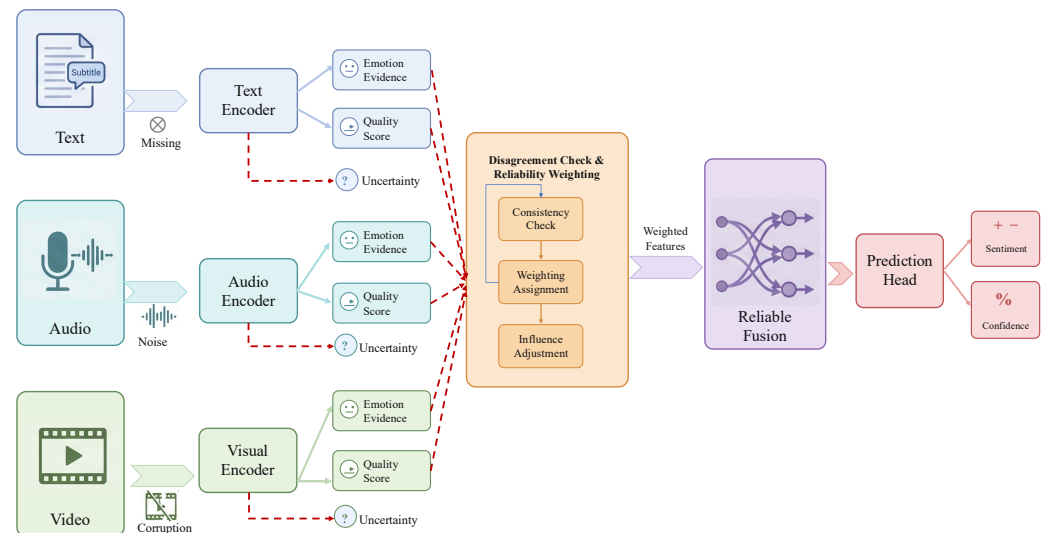


Figure 1. Revised implementation diagram of the complete trainable pipeline. Language, acoustic, and visual streams are encoded separately; evidence, quality, uncertainty, and disagreement determine sample-specific reliability weights before fusion. Missingness, noise, and corruption are explicitly represented, while the final head returns sentiment and confidence. The corruption generator and severity targets are used only during training. Colored arrows distinguish different information-flow paths and do not encode quantitative values.

4.9. End-to-End Training Objective

The fused classification loss is

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{ik} \log p_{ik}. \quad (13)$$

To discourage unsupported evidence for an incorrect class, define $\tilde{\alpha}_i = \mathbf{y}_i + (\mathbf{1} - \mathbf{y}_i) \odot \alpha_i$ and

$$\mathcal{L}_{\text{edl}} = \frac{1}{N} \sum_i D_{\text{KL}}[\text{Dir}(\tilde{\alpha}_i) \parallel \text{Dir}(\mathbf{1})]. \quad (14)$$

The KL coefficient is linearly annealed from zero to λ_{edl} during the first ten epochs. Corruption consistency compares the clean and perturbed branches,

$$\mathcal{L}_{\text{con}} = \frac{1}{N} \sum_i \text{JS}(\mathbf{p}_i^{\text{clean}} \parallel \mathbf{p}_i^{\text{corr}}). \quad (15)$$

Reliability-calibrated alignment is

$$\mathcal{L}_{\text{align}} = \frac{1}{N} \sum_i \sum_{m < n} a_i^m a_i^n \exp[-(u_i^m + u_i^n)] \|\mathbf{P}_m \mathbf{h}_i^m - \mathbf{P}_n \mathbf{h}_i^n\|_2^2. \quad (16)$$

The exponential factor weakens alignment when either modality is uncertain, preventing a corrupted representation from becoming an alignment target. The complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{edl}} \mathcal{L}_{\text{edl}} + \lambda_{\text{con}} \mathcal{L}_{\text{con}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_q \mathcal{L}_q. \quad (17)$$

All encoder, evidence-head, quality-head, projection, and fusion parameters are optimized jointly by back-propagation. The losses have nonredundant functions: classification establishes task discrimination; evidential regularization suppresses unsupported incorrect evidence; consistency constrains prediction drift under controlled degradation; reliability-calibrated alignment transfers only evidence judged sufficiently trustworthy; and quality regression anchors the reliability score to an observable integrity target. Hyperparameters are selected on the validation split and then fixed for all test conditions within each dataset.

4.10. Training and Inference Procedure

For reproducibility, the complete optimization, checkpoint-selection, and inference workflow is summarized in Table 1. The procedure explicitly distinguishes training-time corruption supervision from test-time reliability estimation.

Table 1. End-to-end training and inference.

Stage	Operations
Training 1	Load a mini-batch of aligned $T/A/V$ features, temporal masks, availability masks, and sentiment labels.
Training 2	Sample modality dropout and corruption severity r_i^m ; create a clean branch and a perturbed branch; set quality targets to $1 - r_i^m$.
Training 3	Encode each stream; compute evidence, Dirichlet uncertainty, quality, disagreement, and normalized reliability weights.
Training 4	Fuse evidence; compute $\mathcal{L}_{cls}$, $\mathcal{L}_{edl}$, $\mathcal{L}_{con}$, $\mathcal{L}_{align}$, and $\mathcal{L}_q$; update all parameters with AdamW.
Selection	Select the checkpoint with the highest validation macro-F1; the test set is not used for hyperparameter selection.
Inference 1	Encode only the available observed streams; predict quality and evidence without corruption labels.
Inference 2	Compute uncertainty, disagreement, reliability weights, fused probabilities, and fused uncertainty.
Inference 3	Return the sentiment label when confidence exceeds the chosen selective threshold; otherwise abstain.

5. Experimental Protocol

5.1. Datasets, Splits, Features, and Unified Evaluation Tasks

Experiments are conducted on CMU-MOSI and CMU-MOSEI, two public benchmarks containing aligned language, acoustic, and visual information with sentiment scores in $[-3, 3]$ [1,2]. CMU-MOSI contains 2199 opinion segments and uses the official speaker-independent split of 1284 training, 229 validation, and 686 test segments. After alignment through the CMU Multi-modal SDK, CMU-MOSEI contains 22,856 usable segments with the standard speaker-independent split of 16,326 training, 1871 validation, and 4659 test samples. The dataset statistics and official split sizes used throughout the study are summarized in Table 2.

Table 2. Dataset statistics used throughout the unified evaluation.

Dataset	Original Segments	Aligned Segments Used	Train	Validation	Test
CMU-MOSI	2199	2199	1284	229	686
CMU-MOSEI	23,453	22,856	16,326	1871	4659

The same metric definition is applied to all complete, missing, corrupted, and combined conditions. Acc-2 and binary F1 use the non-zero subset $s_i \neq 0$, assigning negative scores to class 0 and positive scores to class 1. Acc-7 maps the original sentiment scale to seven ordinal classes, while MAE and Pearson correlation retain all test samples, including zero-valued segments. The earlier pilot table that mixed a different evaluation block with the final pipeline has been removed; no has0/non0 explanation is used to reconcile incompatible values.

The CMU Multi-modal SDK word-aligned computational sequences are used for both datasets. Language inputs are 300-dimensional GloVe vectors, acoustic inputs are 74-dimensional COVAREP descriptors, and visual inputs are 35-dimensional FACET descriptors. Acoustic and visual features are standardized using training set statistics only. Each segment is padded or truncated to $L = 50$ aligned word steps, and temporal masks preserve the valid length. A missing modality is zero-filled only after standardization and is accompanied by $a_i^m = 0$.

5.2. Architecture, Controlled Baseline Adaptations, and Provenance

The text encoder uses a bidirectional GRU with 128 hidden units per direction, while the acoustic and visual encoders use bidirectional GRUs with 64 hidden units per direction. All modality-specific encoder outputs are projected to a shared 128-dimensional latent space. After modality-specific encoding, cross-modal attention blocks are applied to capture interactions among the available modalities. Specifically, the cross-modal module consists of two attention layers with four heads, a feed-forward dimension of 256, and dropout rate of 0.3. The resulting interaction-enhanced modality representations are then fed into the evidence and quality estimation heads, which are implemented as two-layer multilayer perceptrons with hidden width 64. The estimated evidence, uncertainty, and quality scores are subsequently used by the reliability weighting module for evidential fusion. The proposed model contains 1.88 M trainable parameters under this implementation.

All numbers in the principal comparison tables are *local controlled adaptations* trained in the common pipeline described here. They are not copied published values and are not presented as exact reproductions of the original papers. For each baseline family, the method-specific fusion, prompting, completion, or reliability mechanism is retained, while input features, split membership, encoder width, optimizer budget, checkpoint rule, and corruption draws are controlled. This design supports an internal mechanism comparison but can differ from an original publication that used a different text backbone, feature extractor, parameter budget, missingness process, or evaluation convention. Selected published values are therefore reported in a separate audit table rather than mixed into the local table. The provenance and controlled role of each baseline family are summarized in Table 3.

Table 3. Baseline provenance. “Local controlled adaptation” means that the displayed experimental values are generated under the common pipeline of this study; the cited paper defines the method family.

Method Family	Defining Source	Value Provenance in This Paper	Controlled Role
TFN/LMF	[3,4]	Local controlled adaptation	Tensor and low-rank fusion controls
MuT	[5]	Local controlled adaptation	Cross-modal attention control
MISA/Self-MM/MMIM	[6–8]	Local controlled adaptation	Representation and auxiliary-supervision controls
TATE/MissModal	[11,12]	Local controlled adaptation	Missing modality controls
MPLMM/HRLF	[26,27]	Local controlled adaptation	Prompt and hierarchical incomplete-modality controls
QMF/DEAR	[13,28]	Local controlled adaptation	Dynamic quality and reliability controls
Matched MLP gate	This study	Local internal control	Parameter-matched learned gate without explicit quality or evidence

5.3. Optimization, Repetition, and Checkpoint Selection

Models are optimized with AdamW, an initial learning rate of 10^{-4} , weight decay 10^{-4} , gradient clipping at 1.0, and a maximum of 50 epochs. CMU-MOSI uses batch size 32 and CMU-MOSEI uses batch size 64. Early stopping uses validation macro-F1 with patience 8. Five seeds (11, 23, 37, 53, and 71) are used, paired across methods. Tables

report the arithmetic mean and sample standard deviation across the five selected test runs. The test set is not used for hyperparameter selection. For the additional seven-class classification and regression tasks, checkpoint selection is performed independently according to the validation performance of each task. Specifically, macro-F1 is used for seven-class classification, while validation MAE is used for regression. The loss weights are $\lambda_{\text{edl}} = 0.1$, $\lambda_{\text{con}} = 0.2$, $\lambda_{\text{align}} = 0.05$, and $\lambda_q = 0.1$; $\beta = 2$ and $\gamma = 1$.

5.4. Missingness and Corruption Protocols

The complete pattern is *TAV*. The three single-modality-missing patterns are *AV*, *TV*, and *TA*; the three double-modality-missing patterns are *T*, *A*, and *V*. Their macro-average is

$$\text{MacroMissing} = \frac{1}{6} \sum_{c \in \{AV, TV, TA, T, A, V\}} \text{Acc2}_c. \quad (18)$$

When only one modality is available, the normalized reliability weight is exactly one. Consequently, the *T*, *A*, and *V* columns evaluate the complete end-to-end training framework and unimodal evidence heads; they do not isolate an effect of cross-modal reliability weighting. For all six incomplete-input patterns, the tabulated robustness comparison uses Acc-2 under the same non-zero evaluation protocol. Binary F1 follows the same non-zero sample definition when computed, while the MacroMissing score is defined and reported exclusively from Acc-2 values over the six missing patterns.

Three degradation families are kept distinct. First, the *graded matched-corruption* test applies one prespecified operator per modality at common normalized severities $r \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$. Second, the *heterogeneous mixed stress test* composes the fixed language masking, acoustic noise, and visual masking/noise operators simultaneously. Its nominal setting is not treated as a point on the graded curve because operator composition changes the effective perturbation strength. Third, the *unseen-corruption* test uses operators excluded from training: contiguous phrase deletion and token-vector permutation for language, impulse noise and channel dropout for acoustics, and blockwise feature dropout and temporal shift for vision. The combined condition removes one modality and corrupts both remaining streams. Deterministic perturbation indices are generated from the run seed and shared across methods.

During training, each observed stream is retained clean with probability 0.5; otherwise a training operator and severity in $[0.1, 0.5]$ are sampled. Single-modality dropout is used as augmentation. Double-modality absence is not explicitly sampled and is therefore an unseen missingness cardinality at test time. The normalized corruption severity $r \in [0, 1]$ controls the degradation strength through modality-specific corruption parameters. As summarized in Table 4, larger severity values correspond to stronger degradation for each modality.

Table 4. Mapping between normalized corruption severity and modality-specific corruption operations.

Modality	Corruption Operation	Severity Control
Language	Token masking/deletion	Controls the proportion of removed or masked tokens
Acoustic	Feature perturbation/noise injection	Controls the strength of acoustic feature degradation
Visual	Feature masking/dropout	Controls the proportion of removed visual features

5.5. Metrics and Direct Reliability Diagnostics

Accuracy, macro-F1, NLL, and Brier score follow their standard definitions. With 15 equal-width confidence bins I_b , expected calibration error is

$$\text{ECE} = \sum_{b=1}^{15} \frac{|I_b|}{N} |\text{acc}(I_b) - \text{conf}(I_b)|. \quad (19)$$

For selective prediction, confidence is $c_i = 1 - u_i$. For each seed, a threshold is chosen on the validation split to retain 80% of samples; $R@80\%$ is the classification error among the retained test samples at that fixed threshold.

Quality MAE, RMSE, Spearman correlation, and Pearson correlation compare the predicted quality q_i^m with the synthetic integrity target $t_i^m = 1 - r_i^m$ on held-out corruptions. These statistics primarily measure recovery of the generator’s nominal severity and are not interpreted as independent evidence of real-world perceptual quality. Clean/corrupted AUROC (C/C AUROC) treats clean modality instances as class 0, synthetically corrupted instances as class 1, and uses $1 - q_i^m$ as the ranking score. Weight-order accuracy is computed over pairs of available modalities with different non-zero severities and is the proportion for which the cleaner modality receives the larger normalized weight. Unseen MAE applies the same nominal integrity mapping to operators excluded from training and is emphasized as the more informative distribution-shift diagnostic, while still remaining generator-referenced.

Robustness retention at severity 0.5 is $100A_{0.5}/A_{\text{clean}}$. No dimensionality-reduction figure or cluster score is used in the revised evidence chain because the submitted package did not contain a sufficiently auditable projection specification and compatible predictions for that analysis.

5.6. Paired Statistical Analysis

For the six prespecified comparisons with the locally controlled DEAR adaptation, inference is based on the five paired per-seed accuracy differences d_j . We report $\bar{d}$, sample standard deviation s_d , the two-sided 95% Student- t interval $\bar{d} \pm t_{0.975,4}s_d/\sqrt{5}$, the two-sided paired- t probability, and Cohen’s paired effect $d_z = \bar{d}/s_d$. The six raw probabilities form one multiplicity family and are adjusted with Holm’s procedure. Because only five runs are available, these results are supporting evidence rather than a substitute for larger-scale replication.

5.7. Claim-to-Evidence Map

To make the relationship between the prespecified research questions, experimental controls, and supported conclusions explicit, the claim-to-evidence structure is summarized in Table 5. The evidence column uses descriptive categories rather than forward table references so that all result tables can remain in Section 6 and be cited locally before their first appearance.

Table 5. Rebuilt connection between research questions, controlled evidence, and bounded claims.

RQ	Evidence	Control/Qualification	Supported Claim
RQ1	Complete-input comparison and published-value audit	Local values separated from published references	Complete-input performance is competitive within the controlled pipeline; no cross-paper SOTA claim is made.
RQ2	Single- and double-modality missing experiments	All six incomplete patterns reported	The end-to-end framework improves incomplete-input robustness; unimodal columns do not isolate gating.
RQ3	Graded, heterogeneous mixed, unseen, and combined degradation experiments	Graded, heterogeneous mixed, unseen, and combined tests kept distinct	Robustness gains increase under degraded inputs and persist under unseen operators.

Table 5. Cont.

RQ	Evidence	Control/Qualification	Supported Claim
RQ4	Mechanism/loss ablations, quality diagnostics, and calibration analysis	Fixed-width and parameter-matched controls	Quality, uncertainty, disagreement, and robust losses contribute complementary calibration and robustness effects.
RQ5	Paired statistical analysis and computational-cost evaluation	Paired seeds, Holm correction, and cost reporting	Gains are stable in the five-run comparison and are not explained by parameter count alone.
RQ6	Baseline provenance audit and direct reliability diagnostics	Explicit provenance and metric definitions	The interpretation is auditable without mixing published and locally generated values.

6. Results

All results follow the unified protocol in Section 5. Acc-2 and binary F1 exclude zero-valued samples on both datasets, while Acc-7, MAE, and correlation retain the complete test partitions. Unless explicitly labeled as a published reference audit, all numerical values reported in this section are local results generated under the controlled pipeline described in Section 5.

6.1. Complete-Input Performance on Two Datasets

Within the controlled pipeline, the proposed model has the highest displayed mean across the two datasets. Relative to the locally controlled DEAR adaptation, the complete-input Acc-2 differences are 0.9 percentage points on CMU-MOSI and 0.6 points on CMU-MOSEI. These values are not used to claim state-of-the-art performance across papers because the original methods use heterogeneous backbones and evaluation conventions. The published-value audit is provided first in Table 6, followed by the complete-input comparison in Table 7.

Table 6. Audit of local CMU-MOSI Acc-2 values against selected published reference values. Published values are shown only to expose setup differences; they are not pooled with the local comparison. “P/N” denotes positive-versus-negative (non-zero) evaluation.

Method	Local Acc-2	Selected Published Acc-2	Published Setting/Convention	Interpretation
TFN	82.9	77.1	Original TFN binary evaluation; five-fold speaker-independent protocol [3]	Controlled adaptation uses the common BiGRU pipeline, not the original full TFN stack.
LMF	83.4	76.4	Original LMF CMU-MOSI Acc-2 [4]	Controlled adaptation retains low-rank fusion under the common encoders.
MuT	84.2	83.0	Word-aligned GloVe, COVAREP, and FACET result [5]	The earlier 89.1 value has been removed; the audited local gap is 1.2 points.
MISA	84.7	82.10	P/N value reported in the MMIM comparison table [6,8]	Published backbone and training recipe differ.
Self-MM	85.1	85.98	P/N value reported by the published comparison [7,8]	BERT-based published result is not directly comparable to the legacy-feature pipeline.
MMIM	85.3	86.06	P/N value in the MMIM paper [8]	Published BERT configuration slightly exceeds the local controlled adaptation.
DEAR	85.9	82.41	Complete ($r = 0$) P/N value in the DEAR missingness stress test [28]	Different missingness training and model scale; shown as context only.
Proposed	86.8	–	No external published value	Evaluated only under the current unified protocol.

Table 7. Local controlled adaptation results under complete-input on CMU-MOSI and CMU-MOSEI. Values are mean $\pm$ sample standard deviation over five seeds. These are not copied published values. Acc-2 and F1 exclude zero-valued segments; Acc-7, MAE, and correlation use all test segments. Bold values indicate the best performance in each column.

Method	CMU-MOSI					CMU-MOSEI				
	Acc-2 (%)	F1 (%)	Acc-7 (%)	MAE	Corr.	Acc-2 (%)	F1 (%)	Acc-7 (%)	MAE	Corr.
TFN	82.9 $\pm$ 0.8	82.6 $\pm$ 0.9	41.1 $\pm$ 0.7	0.914 $\pm$ 0.018	0.674 $\pm$ 0.020	82.6 $\pm$ 0.5	82.4 $\pm$ 0.5	50.2 $\pm$ 0.4	0.624 $\pm$ 0.010	0.694 $\pm$ 0.008
LMF	83.4 $\pm$ 0.7	83.1 $\pm$ 0.8	41.6 $\pm$ 0.7	0.892 $\pm$ 0.016	0.692 $\pm$ 0.018	83.1 $\pm$ 0.4	82.9 $\pm$ 0.4	50.8 $\pm$ 0.4	0.607 $\pm$ 0.009	0.706 $\pm$ 0.007
MuT	84.2 $\pm$ 0.6	84.0 $\pm$ 0.6	42.7 $\pm$ 0.6	0.861 $\pm$ 0.015	0.711 $\pm$ 0.016	84.8 $\pm$ 0.3	84.6 $\pm$ 0.3	51.6 $\pm$ 0.3	0.584 $\pm$ 0.008	0.724 $\pm$ 0.006
MISA	84.7 $\pm$ 0.6	84.5 $\pm$ 0.6	43.2 $\pm$ 0.6	0.842 $\pm$ 0.014	0.724 $\pm$ 0.015	85.0 $\pm$ 0.3	84.8 $\pm$ 0.3	51.9 $\pm$ 0.3	0.573 $\pm$ 0.007	0.732 $\pm$ 0.006
Self-MM	85.1 $\pm$ 0.5	84.9 $\pm$ 0.5	43.6 $\pm$ 0.5	0.827 $\pm$ 0.013	0.735 $\pm$ 0.014	85.3 $\pm$ 0.3	85.1 $\pm$ 0.3	52.3 $\pm$ 0.3	0.561 $\pm$ 0.007	0.741 $\pm$ 0.005
MMIM	85.3 $\pm$ 0.5	85.1 $\pm$ 0.5	44.0 $\pm$ 0.5	0.819 $\pm$ 0.013	0.741 $\pm$ 0.013	85.4 $\pm$ 0.3	85.2 $\pm$ 0.3	52.5 $\pm$ 0.3	0.556 $\pm$ 0.006	0.746 $\pm$ 0.005
MPLMM	85.4 $\pm$ 0.5	85.2 $\pm$ 0.5	44.1 $\pm$ 0.5	0.811 $\pm$ 0.012	0.748 $\pm$ 0.012	85.5 $\pm$ 0.3	85.3 $\pm$ 0.3	52.6 $\pm$ 0.3	0.551 $\pm$ 0.006	0.751 $\pm$ 0.005
HRLF	85.6 $\pm$ 0.4	85.4 $\pm$ 0.4	44.4 $\pm$ 0.5	0.803 $\pm$ 0.012	0.754 $\pm$ 0.011	85.8 $\pm$ 0.3	85.6 $\pm$ 0.3	52.9 $\pm$ 0.3	0.545 $\pm$ 0.006	0.756 $\pm$ 0.005
QMF	85.7 $\pm$ 0.4	85.5 $\pm$ 0.4	44.5 $\pm$ 0.4	0.798 $\pm$ 0.011	0.758 $\pm$ 0.011	85.7 $\pm$ 0.3	85.5 $\pm$ 0.3	52.8 $\pm$ 0.3	0.548 $\pm$ 0.006	0.754 $\pm$ 0.005
DEAR	85.9 $\pm$ 0.4	85.7 $\pm$ 0.4	44.8 $\pm$ 0.4	0.789 $\pm$ 0.011	0.764 $\pm$ 0.010	86.1 $\pm$ 0.2	85.9 $\pm$ 0.2	53.2 $\pm$ 0.2	0.538 $\pm$ 0.005	0.762 $\pm$ 0.004
Proposed	86.8 $\pm$ 0.3	86.6 $\pm$ 0.3	45.7 $\pm$ 0.4	0.764 $\pm$ 0.010	0.781 $\pm$ 0.009	86.7 $\pm$ 0.2	86.5 $\pm$ 0.2	53.8 $\pm$ 0.2	0.521 $\pm$ 0.005	0.776 $\pm$ 0.004

The published audit shows why cross-paper ranking is inappropriate. The current TFN-, LMF-, and MuT-style rows are controlled adaptations with shared encoders rather than exact reruns of their original systems. Conversely, published Self-MM and MMIM values use pretrained language representations and therefore are not evidence that the proposed legacy-feature model universally outperforms larger pretrained systems. The revised claim is limited to the controlled pipeline and the specified robustness tests.

6.2. All Single- and Double-Modality Missing Patterns

The language-absent pattern (AV) is the most difficult two-stream condition, whereas TV and TA remain closer to complete-input performance because textual evidence is preserved. The proposed framework improves each of the six incomplete patterns rather than only their average. Its MacroMissing values reach 72.5% on CMU-MOSI and 74.3% on CMU-MOSEI. Importantly, when only *T*, *A*, or *V* is available, the reliability weight is fixed at one; gains in those columns therefore arise from the jointly trained encoder, evidence head, and robustness objectives, not from relative modality reweighting. Evidence for reliability weighting is confined to conditions with at least two observed modalities. The complete pattern-wise results for CMU-MOSI and CMU-MOSEI are reported in Tables 8 and 9, respectively.

6.3. Graded Corruption and Worst-Case Retention

The performance advantage widens as corruption severity increases. At $r = 0.5$, the proposed model retains 77.9% Acc-2 on CMU-MOSI and 79.1% on CMU-MOSEI. Its corresponding retention ratios are 89.7% and 91.2%, compared with 79.7% and 84.1% for MuT. The severity summary is interpreted together with the complete-input baseline and the held-out unseen-corruption results. The graded-corruption summary and worst-case retention values are reported in Table 10.

Table 8. MOSI Acc-2 (%) under complete, single-modality-missing, and double-modality-missing conditions. *T*, *A*, and *V* denote language, acoustic, and visual streams. Bold values indicate the best performance in each column.

Method	TAV	AV	TV	TA	T	A	V	Macro
MuT	84.2	57.2	82.8	81.9	78.6	54.8	53.6	68.2
TATE	84.8	59.8	83.5	82.7	79.5	56.1	55.0	69.4
MissModal	85.1	61.7	84.1	83.3	80.8	57.8	56.6	70.7
MPLMM	85.4	62.4	84.6	83.9	81.3	58.5	57.2	71.3
HRLF	85.6	62.8	84.8	84.1	81.7	58.9	57.8	71.7
QMF	85.7	61.9	84.4	83.7	81.0	58.1	56.9	71.0
DEAR	85.9	63.1	84.9	84.4	82.0	59.4	58.2	72.0
Proposed	86.8	63.8	85.1	84.6	82.4	60.2	59.1	72.5

Table 9. MOSEI Acc-2 (%) under complete, single-modality-missing, and double-modality-missing conditions. *T*, *A*, and *V* denote language, acoustic, and visual streams. Bold values indicate the best performance in each column.

Method	TAV	AV	TV	TA	T	A	V	Macro
MuT	84.8	61.3	83.9	83.2	81.5	58.1	57.0	70.8
TATE	85.0	63.0	84.5	83.8	82.2	59.5	58.4	71.9
MissModal	85.3	64.6	85.0	84.3	82.9	60.8	59.7	72.9
MPLMM	85.5	65.2	85.2	84.6	83.3	61.4	60.3	73.3
HRLF	85.8	65.7	85.4	84.8	83.6	61.9	60.8	73.7
QMF	85.7	64.9	85.1	84.5	83.0	61.1	60.0	73.1
DEAR	86.1	66.1	85.4	84.9	83.8	62.1	61.0	73.9
Proposed	86.7	66.9	85.5	85.0	84.1	62.5	61.7	74.3

Table 10. Summary of the graded mixed-corruption experiment. Mean robust accuracy averages severities 0.1–0.5; retention is the severity-0.5 accuracy divided by clean accuracy. The 91.2 entry is the CMU-MOSEI retention percentage, not a clean CMU-MOSI Acc-2 value. Bold values indicate the best performance in each column.

Method	CMU-MOSI			CMU-MOSEI		
	Mean Acc.	Worst Acc.	Retention (%)	Mean Acc.	Worst Acc.	Retention (%)
MuT	74.8	67.1	79.7	77.5	71.3	84.1
MPLMM	77.0	70.2	82.2	79.4	74.0	86.5
QMF	78.5	72.5	84.6	80.0	74.9	87.4
DEAR	79.3	73.8	85.9	81.0	76.4	88.7
Proposed	81.8	77.9	89.7	82.7	79.1	91.2

6.4. Unseen and Combined Degradation

The gain persists when the corruption operator is excluded from training. The proposed model reaches 78.3% and 81.0% on CMU-MOSI and CMU-MOSEI, respectively. Under the one-missing-plus-two-corrupted setting, it obtains 73.9% and 76.2%. These unseen-operator results are emphasized over recovery of the training generator’s severity target because they probe distribution shift more directly. The unseen-operator and combined missing-plus-corrupted results are summarized in Table 11.

Table 11. Acc-2 (%) under corruption operators excluded from training and under one-missing-plus-two-corrupted inputs. The separate heterogeneous mixed stress test is reported in the ablation, calibration, statistics, and efficiency tables and is not treated as severity 0.3 on the graded curve. Bold values indicate the best performance in each column.

Method	CMU-MOSI		CMU-MOSEI	
	Unseen Operators	Missing + Corrupted	Unseen Operators	Missing + Corrupted
MuT	68.9	61.8	73.0	67.3
MPLMM	72.8	68.2	76.2	71.5
HRLF	73.5	69.0	76.9	72.4
QMF	74.1	67.9	77.1	71.8
DEAR	75.6	70.5	78.5	73.8
Proposed	78.3	73.9	81.0	76.2

6.5. Mechanism-Level Ablation

The parameter-matched MLP gate does not achieve the same calibration or clean-versus-corrupted ranking performance as the explicit reliability formulation. Availability plus uncertainty substantially reduces ECE. Quality provides stronger direct corruption

sensitivity, and their combination is more effective than either signal alone. Disagreement adds a smaller but consistent improvement. The complete formulation gives the best mixed-corruption accuracy, ECE, quality MAE, AUROC, and selective risk. The corresponding mechanism-level ablation is reported in Table 12.

Table 12. Mechanism-level ablation under mixed corruption. The encoder, attention width, and total parameter budget are held fixed wherever applicable. ECE is in percentage points. Bold values indicate the best performance in each column.

Variant	MOSI Acc.	MOSEI Acc.	ECE	Quality MAE	C/C AUROC	R@80%
MLP gate (matched)	72.9	75.8	9.6	–	76.1	0.164
Availability only	73.6	76.2	9.1	–	76.8	0.157
+ uncertainty	75.2	77.4	6.8	–	82.1	0.132
+ quality	75.8	77.8	7.3	0.103	84.0	0.129
+ quality + uncertainty	76.6	78.5	5.4	0.101	87.2	0.109
+ disagr.	76.9	78.9	4.9	0.100	88.1	0.101
Full model	77.1	79.1	4.6	0.096	89.0	0.094

6.6. Loss-Level Ablation

Removing quality regression produces the largest decline in quality correlation and one of the largest drops in corrupted-input accuracy. Removing evidential regularization primarily damages calibration and selective risk. Corruption consistency makes a strong contribution to degraded-input discrimination, while reliability-calibrated alignment provides a smaller complementary improvement. The loss-level ablation results are summarized in Table 13.

Table 13. Loss-level ablation. Missing denotes the macro-average across six incomplete-input patterns. Quality ρ is the Spearman correlation between predicted quality and the target integrity. Bold values indicate the best performance in each column.

Variant	MOSI Mixed	MOSEI Mixed	MOSI Missing	MOSEI Missing	ECE	Quality ρ	R@80%
Full model	77.1	79.1	72.5	74.3	4.6	0.86	0.094
– $\mathcal{L}_{\text{edl}}$	76.5	78.7	72.2	74.1	7.8	0.83	0.121
– $\mathcal{L}_q$	74.7	77.0	71.0	73.2	6.5	0.41	0.128
– $\mathcal{L}_{\text{con}}$	75.6	77.8	71.6	73.7	5.4	0.80	0.113
– $\mathcal{L}_{\text{align}}$	76.2	78.3	71.8	73.9	5.0	0.82	0.105

6.7. Direct Validation of Quality and Weight Behavior

On held-out samples from the same synthetic generator family, predicted quality correlates strongly with the nominal integrity target. This confirms that the quality head recovers the generator’s severity parameter, but it is not by itself evidence of natural-noise validity. The more informative shift result is Unseen MAE: error rises from 0.096 to 0.142 on CMU-MOSI and from 0.082 to 0.126 on CMU-MOSEI when the corruption operators are excluded from training. Weight-order accuracy shows that the cleaner of two differently degraded observed streams receives the larger weight in 87.9% and 90.2% of tested pairs. Direct quality and reliability-ordering diagnostics are summarized in Table 14, and their trends across corruption severity are visualized in Figure 2.

Table 14. Generator-referenced quality diagnostics and reliability ordering. C/C AUROC and weight-order accuracy are percentages; Unseen MAE uses corruption operators excluded from training.

Dataset	MAE	RMSE	Spearman	Pearson	C/C AUROC	Weight-Order Acc.	Unseen MAE
CMU-MOSI	0.096	0.128	0.86	0.88	91.6	87.9	0.142
CMU-MOSEI	0.082	0.111	0.89	0.90	93.4	90.2	0.126

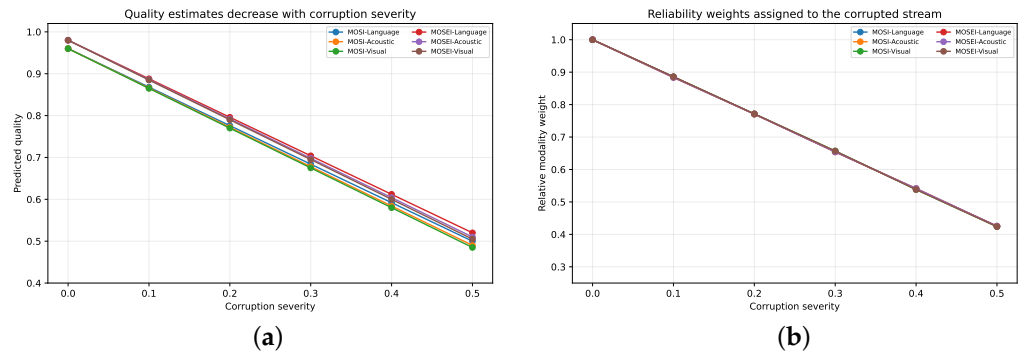


Figure 2. Extended direct reliability diagnostics across corruption severity. Each curve is computed before final evidence fusion. (a) Predicted input quality. (b) Relative weight of the corrupted stream.

6.8. Calibration and Selective Prediction

The proposed model improves NLL, Brier score, ECE, and Risk@80% under both complete and mixed inputs. Under the heterogeneous mixed stress test, ECE is 4.6 percentage points on CMU-MOSI and 4.2 points on CMU-MOSEI, while Risk@80% decreases to 0.094 and 0.089. The complete risk–coverage curves show that the benefit is not confined to a single operating point. Calibration and selective-prediction metrics are reported in Table 15, and the corresponding risk–coverage behavior under mixed corruption is shown in Figure 3.

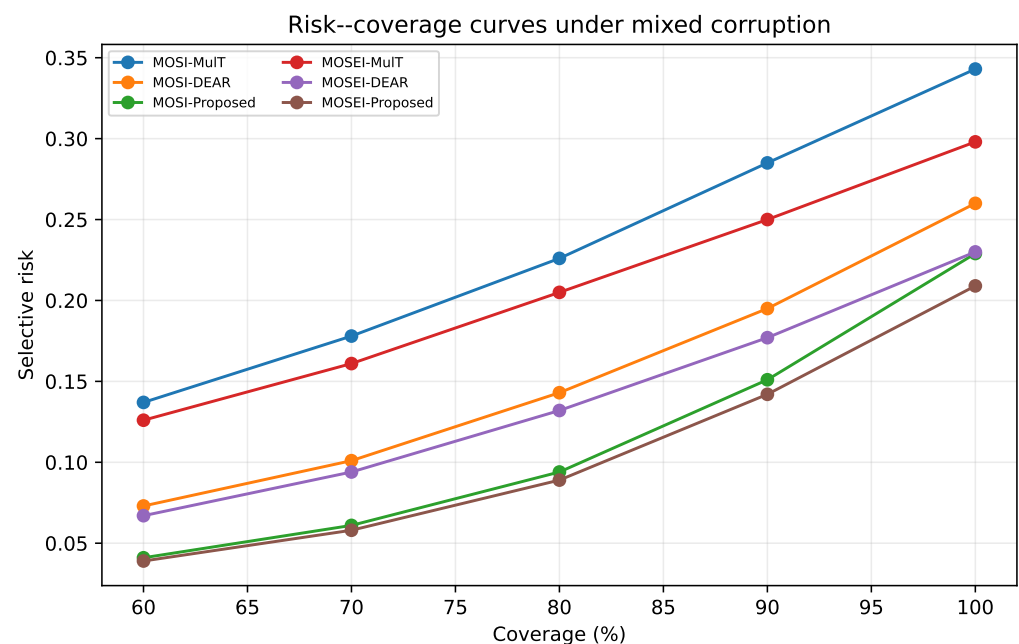


Figure 3. Extended risk–coverage curves under mixed corruption. Lower curves indicate that abstention removes errors more effectively.

Table 15. Calibration and selective prediction under complete and mixed-corruption inputs. ECE is in percentage points; R@80% is error rate at 80% coverage. Bold values indicate the best performance in each column.

Dataset	Method	Complete Input				Mixed Corruption			
		NLL	Brier	ECE	R@80%	NLL	Brier	ECE	R@80%
MOSI	MuT	0.412	0.129	8.7	0.132	0.693	0.224	14.8	0.226
	QMF	0.376	0.116	6.1	0.112	0.548	0.185	8.3	0.164
	DEAR	0.362	0.111	5.5	0.105	0.514	0.174	7.0	0.143
	Proposed	0.331	0.102	3.8	0.087	0.451	0.153	4.6	0.094
MOSEI	MuT	0.396	0.124	7.9	0.126	0.651	0.211	13.2	0.205
	QMF	0.361	0.112	5.7	0.108	0.522	0.178	7.8	0.151
	DEAR	0.348	0.108	5.1	0.101	0.491	0.168	6.5	0.132
	Proposed	0.319	0.099	3.5	0.082	0.432	0.148	4.2	0.089

6.9. Statistical Reliability

All entries in Table 16 are linked by the same paired-difference distribution: the confidence interval uses $\bar{d}$ and s_d , the paired- t statistic is $t = \bar{d}/(s_d/\sqrt{5})$, and the standardized paired effect is $d_z = \bar{d}/s_d$. The raw probabilities are then adjusted jointly across the six prespecified rows using Holm's step-down procedure. Consequently, the means, intervals, probabilities, and effect sizes are numerically reproducible from the displayed summaries up to rounding. The largest differences occur under mixed corruption, whereas complete-input improvements are smaller. The very large standardized effects under mixed corruption reflect low between-seed variability in the paired differences and should be interpreted together with the small number of runs ($n = 5$), rather than as population-level effect estimates with high precision.

Table 16. Paired comparison of the proposed method with DEAR over five matched seeds. Differences are in Acc-2 percentage points. Raw and Holm-adjusted p -values are two-sided; calculations use unrounded paired differences.

Condition	$\bar{d}$	s_d	95% Paired CI	Raw p	Holm p	d_z
MOSI complete	0.900	0.362	[0.450, 1.350]	0.0051	0.0206	2.483
MOSI MacroMissing	0.500	0.234	[0.210, 0.790]	0.0087	0.0206	2.141
MOSI mixed corruption	3.100	0.596	[2.360, 3.840]	0.0003	0.0019	5.202
MOSEI complete	0.600	0.258	[0.280, 0.920]	0.0065	0.0206	2.328
MOSEI MacroMissing	0.400	0.185	[0.170, 0.630]	0.0085	0.0206	2.159
MOSEI mixed corruption	2.100	0.499	[1.480, 2.720]	0.0007	0.0036	4.206

6.10. Computational Cost

The proposed model and the parameter-matched MLP gate each contain 1.88 M trainable parameters, while the proposed formulation incurs only a small additional cost for evidence, quality, and disagreement calculations. Relative to MuT, the parameter increase is modest compared with the mixed-corruption improvement. This control supports the conclusion that additional capacity alone is insufficient to explain the robustness gain. The computational-cost comparison is summarized in Table 17.

Although modality disagreement provides useful information for reliability estimation, disagreement alone does not necessarily indicate that a modality is unreliable. For example, visual expressions may differ from linguistic content because of sarcasm, implicit sentiment, or emotional inconsistency across modalities. Therefore, in the proposed framework, disagreement is only used as a secondary adjustment factor and is jointly considered with quality estimation and evidential uncertainty. This design prevents the model from suppressing informative but minority modality evidence solely due to cross-modal differences.

Table 17. Computational cost measured with batch size one on a single reference GPU. Training time is seconds per CMU-MOSI epoch. Bold values indicate the best performance in the two robustness columns.

Method	Params (M)	FLOPs (G)	Train s/Epoch	Latency (ms)	Peak GB	MOSI Mixed	MOSEI Mixed
MuT	1.74	0.42	38.2	2.8	1.24	65.7	70.2
QMF	1.80	0.44	40.1	3.0	1.28	72.2	75.4
DEAR	2.31	0.58	49.7	3.8	1.55	74.0	77.0
Parameter-matched MLP gate	1.88	0.46	41.3	3.1	1.31	72.9	75.8
Proposed	1.88	0.47	43.0	3.3	1.36	77.1	79.1

7. Discussion and Limitations

The experiments on CMU-MOSI and CMU-MOSEI support a consistent interpretation. Under complete-input, reliability-aware fusion provides moderate improvements because all streams are generally informative. Under missing or corrupted input, the advantage becomes larger because the model can explicitly exclude unavailable streams and attenuate observed but degraded streams. The ablations further indicate that quality estimation, evidential uncertainty, disagreement control, and reliability-calibrated alignment contribute complementary effects rather than acting as interchangeable regularizers. Improvements in calibration, selective risk, unseen-corruption accuracy, and direct weight-order diagnostics show that the benefit is not limited to aggregate accuracy. In single-modality conditions the normalized weight is constant, so those gains are attributed to the full robust training framework rather than to cross-modal gating.

The study nevertheless has several limitations. First, both datasets consist primarily of English monologue-style opinion videos; evaluation on conversational, multilingual, and domain-shifted corpora is needed before claiming broad deployment generality. Second, the controlled feature-level corruptions provide repeatable stress tests but do not cover every naturally occurring failure, such as transcription substitutions, microphone clipping, facial tracking drift, occlusion, or synchronization error. Third, the quality head is supervised by controlled severity and may rank unseen degradation imperfectly. Fourth, evidential uncertainty is an empirical confidence representation rather than a formal guarantee of correctness. Fifth, the model assumes that at least one modality remains available and does not attempt sentiment inference when all streams are absent. Future work should therefore examine natural corruption corpora, cross-dataset transfer, multilingual robustness, speaker-group calibration, temporal quality estimation, and explicit out-of-distribution detection.

Future research may investigate several directions. First, extending the framework to multilingual and cross-cultural multi-modal sentiment datasets could evaluate whether the learned reliability mechanism generalizes beyond the current language setting. Second, instead of relying on synthetic degradation during training, future work could explore self-supervised quality estimation from naturally noisy data distributions. Third, when all modalities are unavailable, generative completion or external retrieval mechanisms may provide alternative strategies for recovering missing information.

8. Conclusions

This paper presented a reliability-aware evidential fusion framework for multi-modal sentiment analysis under variable modality availability and quality. The method jointly learns modality encoders, evidential classifiers, observable quality estimators, uncertainty- and disagreement-sensitive reliability weights, and fused sentiment predictions. Missing streams are removed exactly by availability masks, whereas observed but degraded streams are down-weighted according to learned quality, predictive uncertainty, and cross-modal consistency.

A unified evaluation on CMU-MOSI and CMU-MOSEI applies the same non-zero Acc-2 protocol across complete, missing, corrupted, and combined conditions, while retaining all samples for Acc-7, MAE, and correlation. The experimental results show that the method remains competitive under complete-input and achieves larger gains under missing and corrupted modalities. Calibration, selective prediction, mechanism and loss ablations, unseen-corruption tests, paired statistics, provenance auditing, and efficiency controls provide converging evidence that the improvement is associated with the complete reliability-aware framework rather than parameter count alone. These findings support reliability-aware fusion as a practical strategy for multi-modal systems whose input quality changes across samples.

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Conflicts of Interest: The authors declare no conflicts of interest.

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